

# Results

01 • INDEPENDENT T-TEST

do two groups' means differ?

Table 1. Group statistics

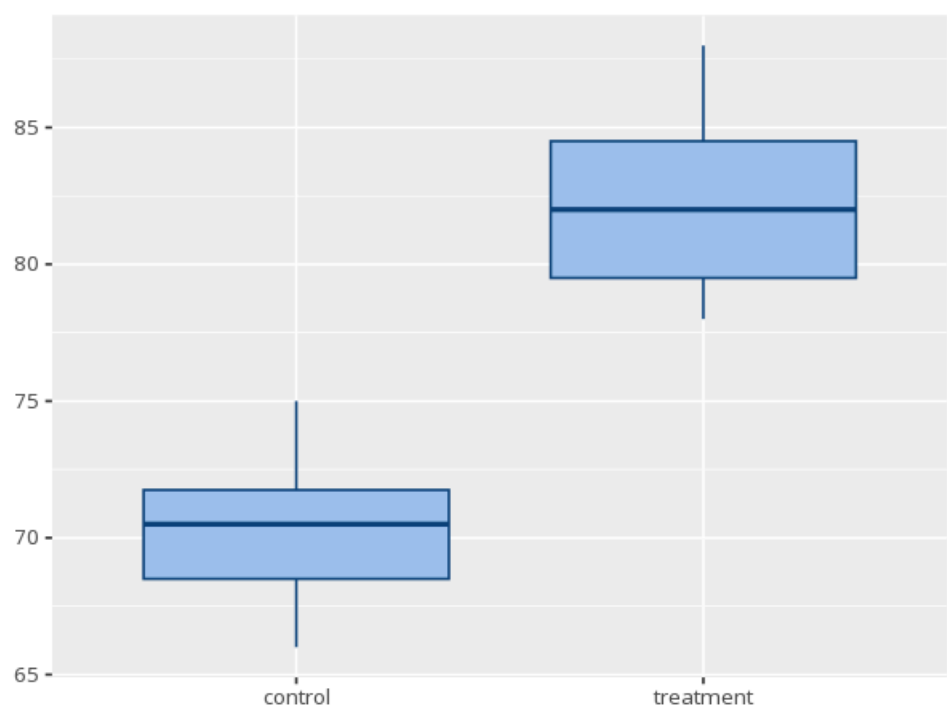
| Group     | N | M     | SD   | SE   |
|-----------|---|-------|------|------|
| control   | 6 | 70.33 | 3.14 | 1.28 |
| treatment | 6 | 82.33 | 3.78 | 1.54 |

Table 2. Independent-samples t-test

| Contrast            | t     | df   | p     | M <sub>diff</sub> | 95% CI          | Cohen's d [95% CI]   |
|---------------------|-------|------|-------|-------------------|-----------------|----------------------|
| control – treatment | -5.98 | 9.68 | <.001 | -12.00            | [-16.49, -7.51] | -3.45 [-5.32, -1.53] |

assumption checks: Levene's test for equal variances & within-group normality (Shapiro-Wilk per group); a Welch row replaces the pooled row when equal-variance is off. (Levene F=0.38, p=.550 · Shapiro control W=0.99, p=.995; Shapiro treatment W=0.96, p=.847 · Welch test)

Figure. Distribution of the outcome by group



How to read this test

Asks whether two groups' averages differ by more than chance. Look at p: below alpha (e.g. 0.05) means significant. The mean difference and 95% CI show the size and precision of the gap; Cohen's d gives effect size (~0.2 small, 0.5 medium, 0.8 large). A large p means no significant difference was detected — not that the group means are equal.

**APA template:** An independent-samples t-test compared control ( $M=70.3$ ,  $SD=3.1$ ) and treatment ( $M=82.3$ ,  $SD=3.8$ ),  $t(9.68)=-5.98$ ,  $p < .001$ ,  $d=-3.45$  [ $-5.32$ ,  $-1.53$ ].

After export · optional feedback — an anonymous, optional satisfaction survey — opens a short Google Form in a new tab

[How did it go? — Share feedback →](#)